
Web Development Fundamentals

Start your web journey with HTML, CSS, and JavaScript. Build real projects from day one.

Level: Beginner | Estimated time: 6 hours | Category: Development

Course overview

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This PDF course pack was generated for the project from publicly available internet resources and the app's built-in lesson outline.

Lesson 1: Introduction to HTML

HTML (HyperText Markup Language) is the foundation of every web page. In this lesson you will learn:

- What HTML is and why it matters
- The basic structure of an HTML document: `<!DOCTYPE>`, `<html>`, `<head>`, `<body>`
- Common HTML tags: headings (h1-h6), paragraphs (p), links (a), images (img)
- Lists: ordered (ol) and unordered (ul)
- Forms and inputs: `<input>`, `<form>`
- Semantic HTML: `<article>`, `<section>`, `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, `<h6>`
- HTML attributes: id, class, href, src, alt
- Nesting elements correctly
- Comments in HTML

Key concept: HTML describes the structure and content of a web page. It is NOT a programming language - it is a markup language.

Lesson 2: CSS Styling Basics

CSS (Cascading Style Sheets) controls the visual presentation of your HTML.

- Selectors: element, class (.), id (#), attribute
- The box model: margin, border, padding, content
- Colors: hex, rgb, hsl, named colors
- Typography: font-family, font-size, font-weight, line-height
- Flexbox: display:flex, justify-content, align-items, gap
- CSS Grid: grid-template-columns, grid-template-rows
- Responsive design with media queries

- CSS Variables: `--my-color: #ff0000`
- Transitions and basic animations
- Pseudo-classes: `:hover`, `:focus`, `:nth-child`

Key concept: CSS cascades - later rules override earlier ones. Specificity determines which rule wins.

Lesson 3: JavaScript Basics

JavaScript adds interactivity and logic to your web pages.

- Variables: `var`, `let`, `const`
- Data types: string, number, boolean, null, undefined, object, array
- Operators: arithmetic (+, -, *, /), comparison (===, !==, <, >), logical (&&, ||, !)
- Control flow: if/else, switch, ternary operator
- Loops: for, while, forEach, for...of
- Functions: declaration, expression, arrow functions
- DOM manipulation: `querySelector`, `getElementById`, `addEventListener`
- Events: click, input, submit, keydown
- Fetch API for HTTP requests
- Async/Await and Promises

Key concept: JavaScript runs in the browser and can modify the page dynamically without reloading.

Further reading from the internet

These official or widely used references match the course topic and can be used for deeper study.

MDN: HTML basics

https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Structuring_content/Basic_HTML_syntax

MDN: CSS first steps

https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Styling_basics

MDN: What is JavaScript?

https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Scripting/What_is_JavaScript